BioPathNet++ for Autoimmune Diseases: Path-Based Transfer Learning for Rare Disease Drug Repurposing

Venkata Naga Sai Yatin Tatikonda*, Srijan Jha, Charan Kumar Reddy Arikeri, Praveen Kumar Yadav, and Priyanka H*

Department of Computer Science and Engineering, PES University, Bengaluru, India
*Corresponding authors: yatin.tvns@gmail.com, drpriyankahsachin@gmail.com

Abstract. Rare autoimmune diseases have few approved drugs, so drug repurposing must rank candidates for diseases whose treatments were never seen in training. This zero-shot setting is studied with BioPathNet++, a BioPathNet-style path-based graph neural network with a channel of four leakage-free prior features, on the PrimeKG knowledge graph. A strict protocol holds out each test disease together with its related diseases and ranks all 7,957 drugs. On 68 unseen test diseases, a known indication appears in the top ten for 76% of diseases, against 57% for the best baseline and 0.8% by chance, and the mean reciprocal rank is significantly higher than the baseline's (0.501 vs. 0.392). When training data are scarce, data-rich autoimmune diseases transfer better than label-matched non-autoimmune diseases; with the full graph they lower accuracy. Ablations show that path reasoning carries the accuracy, while the prior channel mainly stabilizes training across seeds. Registered clinical trials test 13% of the new candidates, against 8% of popular drugs. The output is a ranked list of hypotheses for expert review, not treatment advice.

Keywords: Drug repurposing · Knowledge graphs · Graph neural networks · Zero-shot learning · Transfer learning · Autoimmune diseases

1 Introduction

Repurposing approved drugs, whose safety is known, is faster and cheaper than de novo development [22]. The need is greatest for rare diseases: only 5–7% of the more than 7,000 rare diseases have an FDA-approved drug [15]. Autoimmune disorders affect 10.2% of a 22-million-person UK cohort [9], yet primary biliary cholangitis (PBC), systemic sclerosis (SSc), Sjögren syndrome, autoimmune hepatitis (AIH), myasthenia gravis (MG) and Crohn disease have few disease-specific drugs and little curated data.

Link prediction on knowledge graphs (KGs) of drugs, proteins, phenotypes and diseases ranks candidate drugs [4, 13]. For a disease without known treatments the task is *zero-shot* [15]: knowledge must *transfer* from diseases with known drugs. As autoimmune diseases share immune pathways, the best source

might be the data-rich ones: rheumatoid arthritis (RA), lupus (SLE) and multiple sclerosis (MS). **Aim 1, source of transfer:** must the knowledge come from related autoimmune diseases, or does broad pharmacology suffice? **Aim 2, accuracy:** does a path-based model beat simple rules on unseen diseases? **Aim 3, components:** what do paths, the prior channel, the loss and the graph contribute?

What is new relative to BioPathNet. BioPathNet++ uses the design of BioPathNet [14], a Neural Bellman–Ford network (NBFNet) [33] (Fig. 1); we do not claim a new architecture. Our contributions are (i) a *strict-family* zero-shot protocol that also hides each test disease's relatives, motivated by a leakage audit of our earlier pipeline; (ii) a *controlled transfer study* against label-matched non-autoimmune controls; (iii) a full-graph implementation with memory linear in the number of nodes, plus a stabilizing prior channel; (iv) a head-to-head run of TxGNN's own code, showing that the winner depends on the candidate set; and (v) external checks of new candidates against trials, PubMed and later approvals.

2 Related Work

Biomedical KGs. Hetionet combined 29 public resources and ranked candidates with metapath features [13]. DRKG [16] and PrimeKG [8] followed; PrimeKG separates indications, contraindications and off-label uses for 17,080 diseases. Recent reviews trace KG-based repurposing from feature-based learning to graph neural networks [18, 29]; Table 1 compares the approaches below.

Table 1. Related approaches. Induct.: scores diseases with no known drugs. Hold-out: unit removed for zero-shot testing. Transfer: controlled study of the training source.

Method	Induct.	Explains	Hold-out	Transfer
KGE [6, 27, 31]	no	no	–	no
R-GCN [25]	no	no	–	no
TxGNN [15]	yes	GraphMask	area	no
NBFNet [33]	yes	paths	–	no
BioPathNet [14]	yes	paths	area	no
BioPathNet++	yes	paths	disease+family	controlled

Embeddings and GNNs. KG embeddings [6, 27, 31] and relational GCNs [25] have been used for COVID-19 [19, 32], rare diseases [26] and general repurposing [2, 12]. Their gains depend strongly on dataset construction and evaluation [1, 5], and with one vector learned per node they degrade for diseases without known drugs [15].

Zero-shot and path-based models. TxGNN handles such diseases with disease-similarity metric learning and reports 49.2% better zero-shot indication

prediction than eight other methods [15]. NBFNet scores a (disease, drug) pair by Bellman–Ford message passing from the disease, which is inductive and explainable by paths [33]; BioPathNet adapts it to biomedical KGs and evaluates it on TxGNN's disease-area splits [14]. Recent rare-disease studies instead build disease-specific graphs and explainable GNNs [20, 21].

Leakage. Redundant triples and loose splits overestimate repurposing accuracy [7, 17], and under heavy class imbalance AUROC can look high for weak rankings [24].

3 Data and Protocol

Knowledge graph and labels. We use PrimeKG [8]: 129,375 nodes (7,957 drugs, 17,080 diseases), 4,007,433 edges of 27 non-therapeutic relation types and 42,631 drug–disease pairs of three types: *indication* (the drug is labeled for the disease), *off-label use* and *contraindication*. We call indications and off-label uses *known drugs*, and use "approved" only for regulatory approvals that we checked ourselves. Validation and test positives are indications only. The rare targets use indications plus off-label uses: PBC 1+0, SSc 3+2, Sjögren 2+10, AIH 0+1, MG 4+3, Crohn 16+8. Contraindications are never positives. The targets are case studies; statistics rest on larger held-out sets.

Preprocessing. PrimeKG lists every edge in both directions (8,100,498 rows); we keep one direction per relation, drop duplicates and add explicit inverses (60 relation types). Drug–disease pairs sit in a separate table; a split's message graph receives only those of its training diseases. The path channel uses no node features: nodes differ only by their edges.

Leakage-free zero-shot protocol. A held-out disease is removed with its *family*: its disease–disease neighbors and, for targets, the subtypes that TxGNN's name-based grouping assigns to it. Without this rule, all 16 Crohn indications stayed reachable through "Crohn's colitis". No drug edge of a held-out family enters the message graph, and each training batch hides its own diseases' drug edges, mimicking zero-shot inference. Our earlier pipeline broke these rules: test pairs were graph edges, and a degree score with no learning reached AUROC 0.972. The six target families and 68 random validation diseases are held out; 1,206 diseases with indications remain for training.

Transfer conditions (Aim 1). The test set E holds the 12 diseases with indications in TxGNN's autoimmune area (descendants of DOID:417 [15]) that are neither targets nor RA/SLE/MS relatives. E and the targets are held out in every condition: **A** trains on the RA/SLE/MS families only (11 diseases, 161 indications); $\mathbf{A}'$ on three random sets of 11 non-autoimmune diseases, each matched to A disease by disease on number of indications; **D** on all 1,194 non-autoimmune diseases; $\mathbf{C}_E$ on all 1,185 diseases except E. A vs. A′ asks whether autoimmune data helps when data is scarce, C_E vs. D whether it helps at scale.

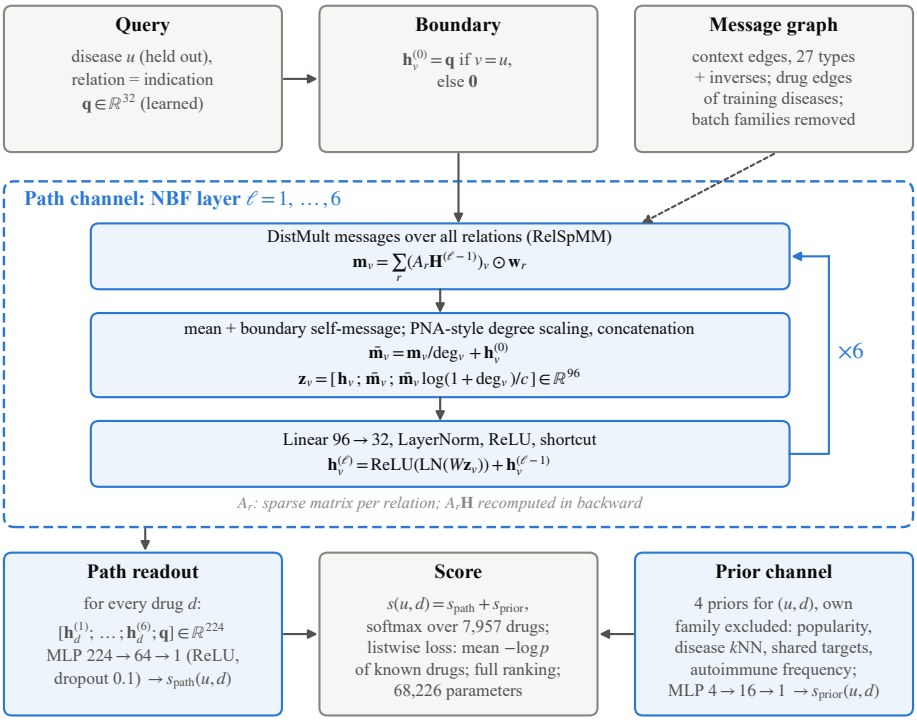


Fig. 1. BioPathNet++ architecture: six NBF layers pass DistMult messages from the query disease u over a graph without the drug edges of u's family; path and prior scores are summed to rank all 7,957 drugs.

4 Method

Path channel. For a query disease u, node states start from an indicator, $h_v^{(0)} = \mathbf{1}[v = u]\, q$, with a learned query vector q, and pass through $L = 6$ layers

$$m_v^{(t)} = \frac{1}{\deg v} \sum_{(x,r,v)} h_x^{(t-1)} \odot w_r^{(t)} \; + \; \mathbf{1}[v = u]\, q, \tag{1}$$

$$h_v^{(t)} = h_v^{(t-1)} + \mathrm{ReLU}\big(\mathrm{LN}\big(W^{(t)}[h_v^{(t-1)};\, m_v^{(t)};\, a_v m_v^{(t)}]\big)\big), \tag{2}$$

with DistMult relation vectors w_r in (1) and a PNA-style degree scaler a_v [10] in (2). An MLP over $[h_d^{(1)}; \ldots; h_d^{(L)}; q]$ scores every drug d in one pass (Fig. 1). One sparse matrix per relation and a custom autograd function that recomputes products in the backward pass make memory grow with nodes, not edges.

Prior channel and training. A small MLP over four features adds a second score: drug popularity among training diseases, a disease-similarity kNN score ($k = 20$, TF–IDF over disease proteins and phenotypes), shared drug-target/disease-gene count, and frequency among autoimmune training diseases,

each computed with the query's family excluded. The loss is the mean negative log-softmax of each disease's known drugs over all drugs. Hyperparameters were fixed in advance, not tuned.

Evaluation. All drugs are ranked with *filtered* ranks (other known drugs removed) [6]. Table 2 defines the metrics. On the test split, MRR [28] is 0.501; the median known-drug rank is 2, and 43% are ranked first. AUROC is high even for popularity (0.830), so it is never used alone [24]. The 68 validation diseases select the stopping epoch; 68 further *test* diseases are used for neither training nor stopping. Of six non-neural baselines and three KG embeddings (TransE, DistMult, RotatE, same graph), the strongest is disease kNN (drugs of the 20 most similar training diseases). CIs are bootstraps over diseases [11]; a difference is significant only if the paired CI excludes zero *and* a two-sided Wilcoxon test gives $p < 0.05$ [30].

Reproducibility. Data: PrimeKG (Harvard Dataverse, https://doi.org/10.7910/DVN/IXA7BM) and TxGNN's area files and code (GitHub, commit 0e0203c). Split seeds: validation 0, test 1, controls 100–102 (JSON). Model: as in Fig. 1 (dimension 32, 6 layers, dropout 0.1). Training: Adam, learning rate 5×10^{-3}, 8 diseases per batch, gradient clipping 1.0, $\leq$20 epochs, patience 5–6 on validation MRR (transfer runs: $\leq$60 epochs, patience 10); seeds 0–2; one NVIDIA T4 per run (5.8 GB peak). Statistics: 1,000 bootstrap resamples (5,000 for paired tests). Code, splits, logs and number-generating scripts: https://github.com/SaiYatin/Capstone_Drug_Repurposing_Project (branch `bpp-v3`).

5 Results

5.1 Aim 1: Transfer to Rare Autoimmune Diseases

Scarce data (Fig. 2). Trained only on the RA/SLE/MS families (A), BioPathNet++ reaches MRR 0.290 on E, against 0.126 with label-matched non-autoimmune diseases (A′): a significant difference of +0.164 (CI +0.037 to +0.316; p=0.016; better on 9 of 12), also with final checkpoints (p=0.021). The targets (0.124 vs. 0.041; p=0.062, $n = 6$) and disease kNN (0.258 vs. 0.086) point the same way. What transfers is a drug-class prior: the A model ranks the same corticosteroids first for every target. With paths only, both stay near the floor (0.007 vs. 0.010), so the advantage comes from the prior channel.

Full scale. Training on all non-autoimmune diseases (D) reaches 0.532 on E; adding the autoimmune diseases (C_E, 3 seeds) lowers it to 0.410 (p=0.010; same validation set), and disease kNN drops too (0.318 to 0.271). On the targets, removing all autoimmune diseases changes nothing (0.487 vs. 0.475; p=0.812).

5.2 Aim 2: Accuracy on Unseen Diseases

On the selection-free test split (Table 3), a known indication is in the top 10 for 76% of diseases, against 57% for disease kNN, 19% for popularity and 0.8% by chance. MRR is 0.501 vs. 0.392 (difference +0.109, CI +0.044 to +0.180;

Table 2. Metrics (per disease over all 7,957 drugs, then averaged) and their value for a random ranking.

Metric	Meaning	Random
Any@10	share of diseases with a known drug in the top 10	0.8%
H@k	share of known drugs in the top k (Hits@k)	k/7957
MRR	mean of 1/rank over known drugs (1 = all first)	≈0.001
AUROC	chance a known drug outranks a random other drug	0.5
AUPRC	area under precision–recall; rewards known drugs at the top	≈0.001

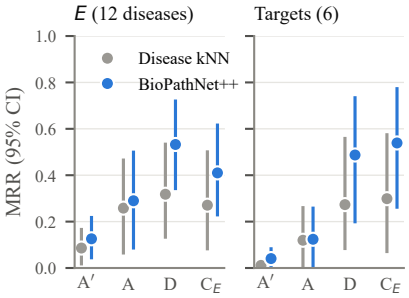


Fig. 2. Aim 1: MRR on E (left) and the targets (right) by training set (95% CIs).

p =0.003). On validation the gain over kNN is +0.113 (p <0.001). BioPathNet++ thus outperforms every baseline evaluated on the same leakage-free split; on the six targets it is ahead for 4 of 6 (not significant).

Table 3. Aim 2: ranking of all 7,957 drugs (metrics: Table 2). AUROC hard/deg.: against drugs indicated for autoimmune diseases / degree-matched drugs. Test: never used for model selection; seed means (3 seeds; KG embeddings 3).

Method	Any@10	H@1	H@10	H@50	AUROC	AUROC hard	AUROC deg.	AUPRC	MRR	Mean rank
Test (68 diseases)										
Random	1%	0.000	0.005	0.006	0.453	0.471	0.500	0.002	0.001	4,349
Popularity	19%	0.042	0.070	0.267	0.885	0.562	0.812	0.084	0.062	918
TransE	27%	0.019	0.061	0.146	0.880	0.674	0.773	0.050	0.037	952
DistMult	22%	0.025	0.089	0.193	0.852	0.638	0.732	0.060	0.047	1,175
RotatE	27%	0.052	0.094	0.209	0.878	0.723	0.785	0.089	0.071	971
Disease kNN	57%	0.347	0.459	0.583	0.907	0.721	0.859	0.429	0.392	742
BioPathNet++	76%	0.435	0.637	0.727	0.959	0.862	0.924	0.556	0.501	329
Validation (68 diseases)										
Random	1%	0.000	0.002	0.004	0.471	0.488	0.490	0.002	0.001	4,205
Popularity	15%	0.034	0.049	0.188	0.830	0.476	0.760	0.069	0.046	1,355
Disease kNN	44%	0.239	0.350	0.490	0.860	0.661	0.822	0.313	0.281	1,118
BioPathNet++	56%	0.353	0.456	0.601	0.938	0.799	0.886	0.424	0.394	495
Rare targets (6 diseases)										
Random	0%	0.000	0.000	0.000	0.476	0.488	0.389	0.002	0.001	4,165
Popularity	50%	0.062	0.148	0.162	0.905	0.460	0.766	0.116	0.099	755
Disease kNN	67%	0.243	0.424	0.725	0.962	0.778	0.917	0.407	0.296	299
BioPathNet++	83%	0.396	0.648	0.695	0.982	0.892	0.954	0.594	0.475	141

5.3 Aim 3: Contribution of Each Component

Paths carry the signal: on validation, paths alone reach MRR 0.405, tied with the full model's 0.394 (p =0.148); the prior channel alone reaches 0.306. These

are not comparable with the test MRR of 0.501: the test diseases are easier for every method (kNN 0.392 on test vs. 0.281 on validation). The prior channel does not raise the mean but cuts seed variance on the targets (sd 0.017 vs. 0.087). Self-adversarial BCE is not significantly worse than the listwise loss (0.378 vs. 0.394; $p = 0.344$), and dropping tissue-expression and drug–drug edges removes 72% of edges at unchanged accuracy ($p = 0.299$).

5.4 Case Studies and Validation

Known drugs. Ursodeoxycholic acid ranks first for PBC in every seed; the four labeled cholinesterase inhibitors hold ranks 1–4 for MG. AIH scores near zero only because PrimeKG lists one off-label drug; the top choice, prednisolone, is standard therapy (comparator in NCT00838214). *Unlabeled candidates* (Table 4). We searched ClinicalTrials.gov for each candidate and two reference sets (popularity top 20; 20 random drugs per target), requiring a whole-name match to a named intervention (A: phase $\geq$3, B: 2, C: 1/observational). A trial *tests* the drug only if its title names it or the drug is in some but not all arms (not as background therapy). Of the model's candidates, 13% have such a trial (9 at phase $\geq$3), against 8% (popularity) and 0% (random); e.g. amifampridine–MG (NCT03304054), ciprofloxacin–Crohn (NCT00609973), both checked by hand. A trial is no proof of benefit: amifampridine helped in MuSK-antibody MG [3] but not in AChR-antibody MG [23]. Ustekinumab–Crohn is a PrimeKG label gap (approved 2016). AI-assisted PubMed labels were each re-checked against the abstract. PubMed shows benefit for 15% of candidates (popularity 10%, random 0%) and harm or no benefit for 5% (popularity 12%): three corticosteroid pairs and N-acetylcysteine for SSc; popularity's are mostly antibiotics that can worsen MG or induce hepatitis.

Temporal check. Six drugs approved after the PrimeKG snapshot (three for Crohn, two for MG, one for PBC) rank at a median of 1,104 of 7,957 (5 of 6 above chance; $p = 0.109$). *Explanations and bias.* Gradient-based path extraction [33] shows that amifampridine reaches MG through the related Lambert–Eaton syndrome and exposes weak links such as ciprofloxacin reaching Crohn through "mouth disease". Corticosteroids fill 41% and antibacterials 21% of top-20 slots, so the lists need expert filtering.

6 Discussion and Limitations

BioPathNet reports MRR of about 0.2–0.8 on TxGNN's disease areas [14], but on other splits; we therefore ran TxGNN's own code on its autoimmune split, training our model only on edges TxGNN also trains on (3 seeds, 13 diseases). Ranking all drugs, BioPathNet++ is significantly better (MRR 0.273 vs. 0.048, $p = 0.040$): only 23% of TxGNN's top-100 drugs have any indication, against 99% for ours. Ranking only drugs with indications, as TxGNN does, nothing is significant (Recall@100 0.614 vs. 0.536; AUROC 0.777 vs. 0.901). BioPathNet was not re-run. Limitations: (i) the targets have 1–24 labeled drugs, so their

Table 4. Evidence for unlabeled top-20 candidates (Ours n=86; others n=120).

Clinical trials	Ours	Pop.	Rand.	Literature and targets	Ours	Pop.	Rand.
Trial names drug, A–C	17%	14%	0%	PubMed co-mention	63%	72%	18%
Trial names drug, A–B	15%	12%	0%	PubMed: benefit	15%	10%	0%
Trial tests drug, A–C	13%	8%	0%	PubMed: harm/no benefit	5%	12%	0%
Trial tests drug, A–B	12%	8%	0%	Drug targets a disease gene	17%	16%	3%

metrics are noisy lower bounds (AIH); (ii) validation diseases also choose the stopping epoch (0.002–0.016 MRR of optimism), which the test split avoids; (iii) E has 12 diseases, and 11-disease models are seed-unstable; (iv) no clinician has reviewed the literature labels.

7 Conclusion

Path-based zero-shot transfer works for rare autoimmune diseases the model never sees: related autoimmune diseases are the best source when data is scarce, the broad pharmacological graph at full scale. BioPathNet++ outperforms every baseline evaluated on the same leakage-free split, recovers established therapies, and ranks later-approved drugs high but not significantly above chance. Paths drive accuracy; the prior channel adds stability.

Acknowledgments. Generative AI tools (Anthropic's Claude) assisted with code, data analysis, and drafting and editing of the text; the authors designed the study, ran all experiments and verified every result against the run logs.

References

1. Ali, M., Berrendorf, M., Hoyt, C.T., Vermue, L., Galkin, M., Sharifzadeh, S., Fischer, A., Tresp, V., Lehmann, J.: Bringing light into the dark: A large-scale evaluation of knowledge graph embedding models under a unified framework. IEEE Transactions on Pattern Analysis and Machine Intelligence **44**(12), 8825–8845 (2022). DOI 10.1109/TPAMI.2021.3124805
2. Bang, D., Lim, S., Lee, S., Kim, S.: Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers. Nature Communications **14**, 3570 (2023). DOI 10.1038/s41467-023-39301-y
3. Bonanno, S., Pasanisi, M.B., Frangiamore, R., Maggi, L., Antozzi, C., Andreetta, F., Campanella, A., Brenna, G., Cottini, L., Mantegazza, R.: Amifampridine phosphate in the treatment of muscle-specific kinase myasthenia gravis: a phase IIb, randomized, double-blind, placebo-controlled, double crossover study. SAGE Open Medicine **6**, 2050312118819013 (2018). DOI 10.1177/2050312118819013
4. Bonner, S., Barrett, I.P., Ye, C., Swiers, R., Engkvist, O., Bender, A., Hoyt, C.T., Hamilton, W.L.: A review of biomedical datasets relating to drug discovery: a knowledge graph perspective. Briefings in Bioinformatics **23**(6), bbac404 (2022). DOI 10.1093/bib/bbac404

5. Bonner, S., Barrett, I.P., Ye, C., Swiers, R., Engkvist, O., Hoyt, C.T., Hamilton, W.L.: Understanding the performance of knowledge graph embeddings in drug discovery. Artificial Intelligence in the Life Sciences **2**, 100036 (2022). DOI 10.1016/j.ailsci.2022.100036
6. Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., Yakhnenko, O.: Translating embeddings for modeling multi-relational data. In: Advances in Neural Information Processing Systems (NeurIPS) (2013)
7. Brière, G., Stosskopf, T., Loire, B., Baudot, A.: Benchmarking the impact of data leakage on the performance of knowledge graph embedding models for biomedical link prediction. Bioinformatics **42**(8), btag608 (2026). DOI 10.1093/bioinformatics/btag608
8. Chandak, P., Huang, K., Zitnik, M.: Building a knowledge graph to enable precision medicine. Scientific Data **10**, 67 (2023). DOI 10.1038/s41597-023-01960-3
9. Conrad, N., Misra, S., Verbakel, J.Y., Verbeke, G., Molenberghs, G., et al.: Incidence, prevalence, and co-occurrence of autoimmune disorders over time and by age, sex, and socioeconomic status: a population-based cohort study of 22 million individuals in the UK. The Lancet **401**(10391), 1878–1890 (2023). DOI 10.1016/S0140-6736(23)00457-9
10. Corso, G., Cavalleri, L., Beaini, D., Liò, P., Veličković, P.: Principal neighbourhood aggregation for graph nets. In: Advances in Neural Information Processing Systems (NeurIPS) (2020). ArXiv:2004.05718
11. Efron, B., Tibshirani, R.J.: An Introduction to the Bootstrap. Chapman & Hall, New York (1993)
12. Gao, Z., Ding, P., Xu, R.: KG-Predict: A knowledge graph computational framework for drug repurposing. Journal of Biomedical Informatics **132**, 104133 (2022). DOI 10.1016/j.jbi.2022.104133
13. Himmelstein, D.S., Lizee, A., Hessler, C., Brueggeman, L., Chen, S.L., Hadley, D., Green, A., Khankhanian, P., Baranzini, S.E.: Systematic integration of biomedical knowledge prioritizes drugs for repurposing. eLife **6**, e26726 (2017). DOI 10.7554/eLife.26726
14. Hu, E.Y., Oleshko, S., Firmani, S., Cheng, H.G., Zhu, Z., Wörheide, M.A., Arnold, M., Colomé-Tatché, M., Tang, J., Xhonneux, S., Marsico, A.: Enhancing link prediction in biomedical knowledge graphs with BioPathNet. Nature Biomedical Engineering **10**(8), 1747–1769 (2026). DOI 10.1038/s41551-025-01598-z
15. Huang, K., Chandak, P., Wang, Q., Havaldar, S., Vaid, A., Leskovec, J., Nadkarni, G.N., Glicksberg, B.S., Gehlenborg, N., Zitnik, M.: A foundation model for clinician-centered drug repurposing. Nature Medicine **30**(12), 3601–3613 (2024). DOI 10.1038/s41591-024-03233-x
16. Ioannidis, V.N., Song, X., Manchanda, S., Li, M., Pan, X., Zheng, D., Ning, X., Zeng, X., Karypis, G.: DRKG – drug repurposing knowledge graph for COVID-19. https://github.com/gnn4dr/DRKG (2020)
17. Kapoor, S., Narayanan, A.: Leakage and the reproducibility crisis in machine-learning-based science. Patterns **4**(9), 100804 (2023). DOI 10.1016/j.patter.2023.100804
18. Kumar, A., Bhandary, S., Hegde, S.G., Chatterjee, J.: Knowledge graph applications and multi-relation learning for drug repurposing: A scoping review. Computational Biology and Chemistry **115**, 108364 (2025). DOI 10.1016/j.compbiolchem.2025.108364
19. Morselli Gysi, D., do Valle, Í., Zitnik, M., Ameli, A., Gan, X., et al.: Network medicine framework for identifying drug-repurposing opportunities for COVID-19.

Proceedings of the National Academy of Sciences **118**(19), e2025581118 (2021). DOI 10.1073/pnas.2025581118
20. Papadimas, F., Svolou, S., Bougiatiotis, K., Aisopos, F., Krithara, A., et al.: Identifying drug repurposing candidates for rare neuro-muscular disorders, using different AI methods on the literature knowledge graph. JAMIA Open **9**(2) (2026). DOI 10.1093/jamiaopen/ooag050
21. Perdomo-Quinteiro, P., Wolstencroft, K., Roos, M., Queralt-Rosiñach, N.: Knowledge graphs and explainable artificial intelligence (AI) for drug repurposing on rare diseases. Neurosymbolic Artificial Intelligence **2** (2026). DOI 10.1177/2949873226 1443101
22. Pushpakom, S., Iorio, F., Eyers, P.A., Escott, K.J., Hopper, S., Wells, A., et al.: Drug repurposing: progress, challenges and recommendations. Nature Reviews Drug Discovery **18**(1), 41–58 (2019). DOI 10.1038/nrd.2018.168
23. Remijn-Nelissen, L., Bakker, W.R., van Gelder, T., Moes, D.J.A.R., Ruiter, A.M., Campman, Y.J.M., Badrising, U.A., Straathof, C.S.M., Verschuuren, J.J.G.M., Tannemaat, M.R.: Efficacy and safety of amifampridine in myasthenia gravis: A randomized, double-blind, placebo-controlled crossover trial. Neurology **106**(8), e214715 (2026). DOI 10.1212/WNL.0000000000214715
24. Saito, T., Rehmsmeier, M.: The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE **10**(3), e0118432 (2015). DOI 10.1371/journal.pone.0118432
25. Schlichtkrull, M., Kipf, T.N., Bloem, P., van den Berg, R., Titov, I., Welling, M.: Modeling relational data with graph convolutional networks. In: The Semantic Web (ESWC), *Lecture Notes in Computer Science*, vol. 10843, pp. 593–607 (2018). DOI 10.1007/978-3-319-93417-4_38
26. Sosa, D.N., Derry, A., Guo, M., Wei, E., Brinton, C., Altman, R.B.: A literature-based knowledge graph embedding method for identifying drug repurposing opportunities in rare diseases. Pacific Symposium on Biocomputing **25**, 463–474 (2020). DOI 10.1142/9789811215636_0041
27. Sun, Z., Deng, Z.H., Nie, J.Y., Tang, J.: RotatE: Knowledge graph embedding by relational rotation in complex space. In: International Conference on Learning Representations (ICLR) (2019). ArXiv:1902.10197
28. Voorhees, E.M.: The TREC-8 question answering track report. In: Proceedings of the Eighth Text REtrieval Conference (TREC-8) (1999)
29. Wei, S., Sasi, C., Piepenbrock, J., Huynen, M.A., 't Hoen, P.A.C.: The use of knowledge graphs for drug repurposing: From classical machine learning algorithms to graph neural networks. Computers in Biology and Medicine **196**, 110873 (2025). DOI 10.1016/j.compbiomed.2025.110873
30. Wilcoxon, F.: Individual comparisons by ranking methods. Biometrics Bulletin **1**(6), 80–83 (1945). DOI 10.2307/3001968
31. Yang, B., Yih, W.t., He, X., Gao, J., Deng, L.: Embedding entities and relations for learning and inference in knowledge bases. In: International Conference on Learning Representations (ICLR) (2015). ArXiv:1412.6575
32. Zhang, R., Hristovski, D., Schutte, D., Kastrin, A., Fiszman, M., Kilicoglu, H.: Drug repurposing for COVID-19 via knowledge graph completion. Journal of Biomedical Informatics **115**, 103696 (2021). DOI 10.1016/j.jbi.2021.103696
33. Zhu, Z., Zhang, Z., Xhonneux, L.P., Tang, J.: Neural Bellman-Ford networks: A general graph neural network framework for link prediction. In: Advances in Neural Information Processing Systems (NeurIPS) (2021). ArXiv:2106.06935